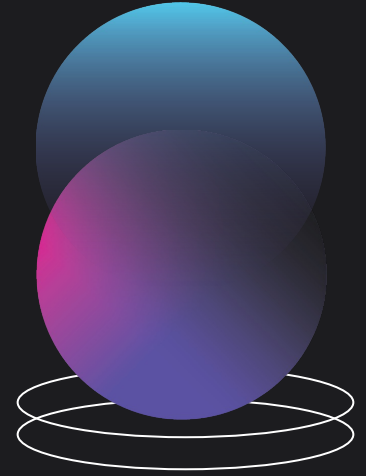
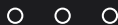


Data Analysis

Predicting Cardiac Pathologies in Children

Bruno Moreira
Tiago Gonçalves





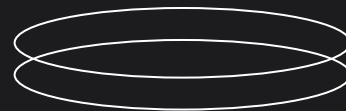
Introduction

This project aims to Apply feature importance methods to rank variables predictive of cardiac pathologies.

Methodology:

- Data Quality Assessment
- Preprocessing (Cleaning, Normalization)
- Statistical Analysis (Bivariate & Multivariate)
- Predictive Modeling (Decision Trees, SVM, Random Forest)

Can we predict cardiac disease using only clinical history, or is the physical exam (murmurs) mandatory?



Data Understanding (Raw Data)

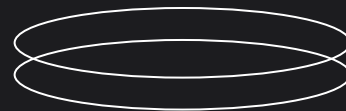
Dataset: Clinical records from Real Hospital Português (Brazil).

Variable Types:

- Numerical: Age, Weight, Height, Heart Rate.
- Nominal/Categorical: Sex, Murmur Type (SOPRO), B2 Sound.

Initial Audit:

- Numerical inconsistencies of various kinds observed.
- Identified inconsistencies in categorical labels (e.g., gender variants).
- Found missing values in the critical target variable (Pathology).



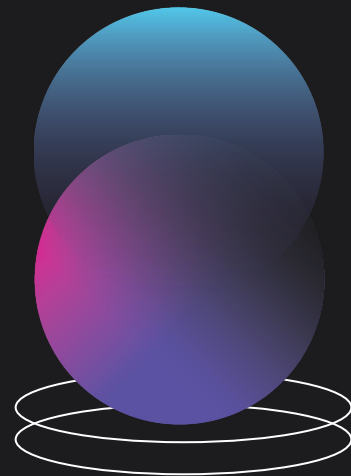
Preprocessing I - Cleaning & Outliers

Data Cleaning Actions:

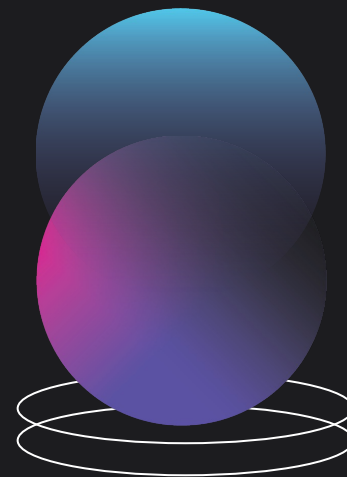
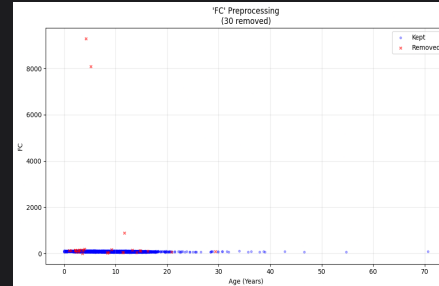
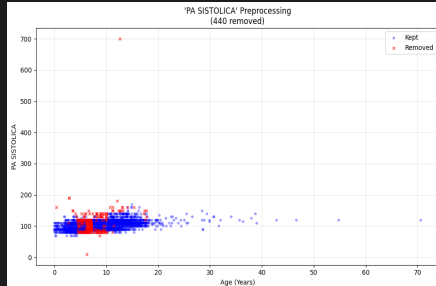
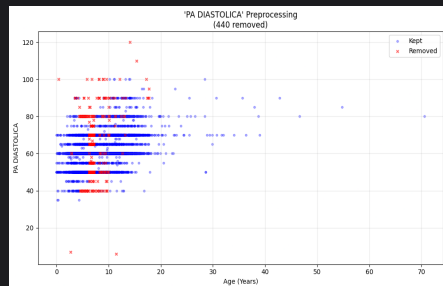
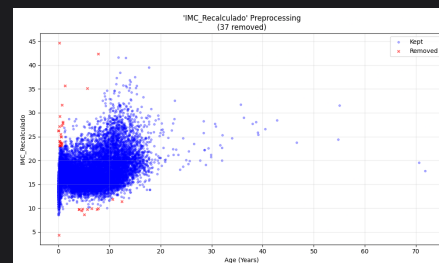
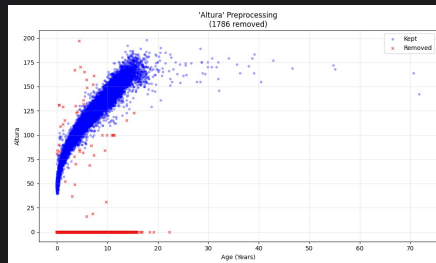
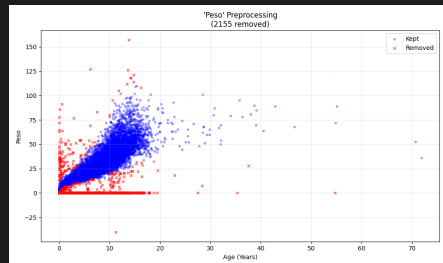
- Removed rows with NaN in target (Supervised Learning requirement).
- Standardized text fields (e.g., 'M' vs 'Masculino').
- Checked for consistencies among values (e.g. 'Age' had to be consistent with 'DN' and 'Atendimento')

Outlier Management:

- Used Boxplots and Interquartile Range (IQR).
- Removed physiological impossibilities (e.g., extreme weights for age) to prevent skewing the normalization process.



Visualization of Data Cleaning Stages

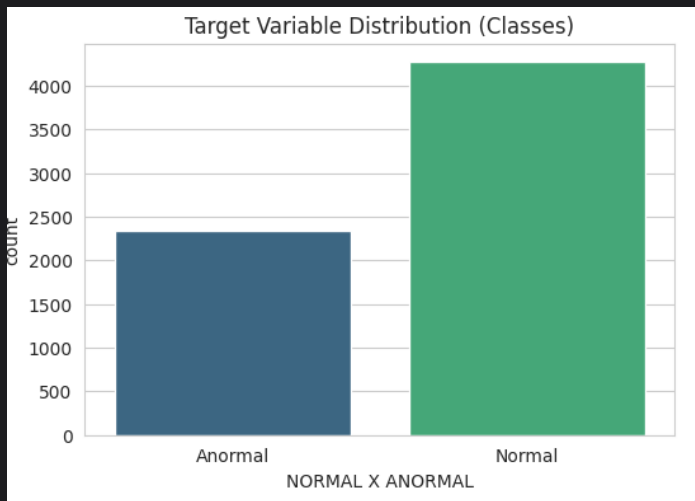


Preprocessing II - Transformation & Scaling

Normalization:

- Applied StandardScaler to all numerical inputs.
- Essential for distance-based algorithms like SVM and K-Means.

Basic Description - Class Balance



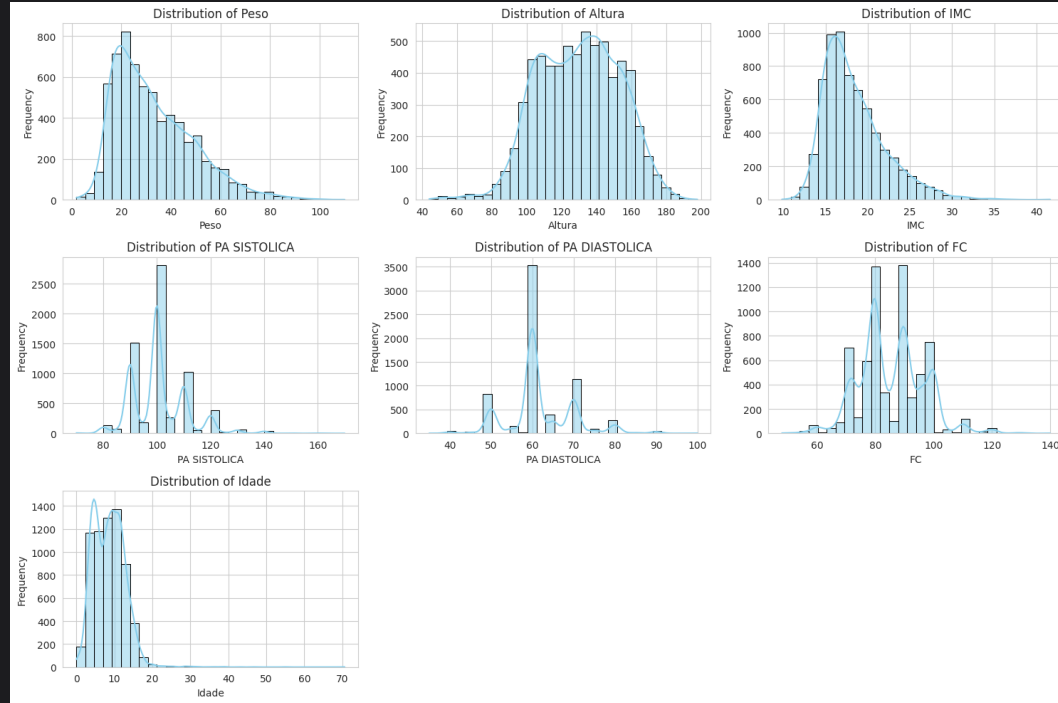
Target Distribution:

Normal: 64.6% (Majority Class)

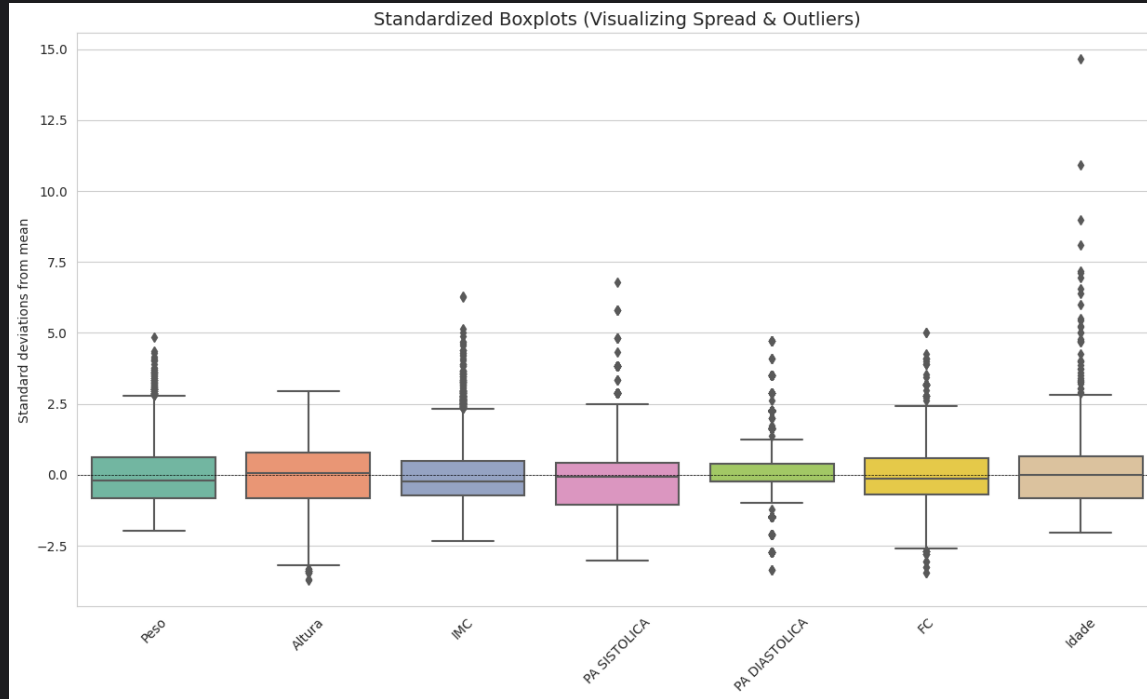
Anormal: 35.4% (Minority Class)

- Naive prediction (Majority Class) yields ~65% Accuracy.
- Standard accuracy metrics are insufficient for performance evaluation.
- Prioritize the minimization of False Negatives (missed diagnoses) over global accuracy.

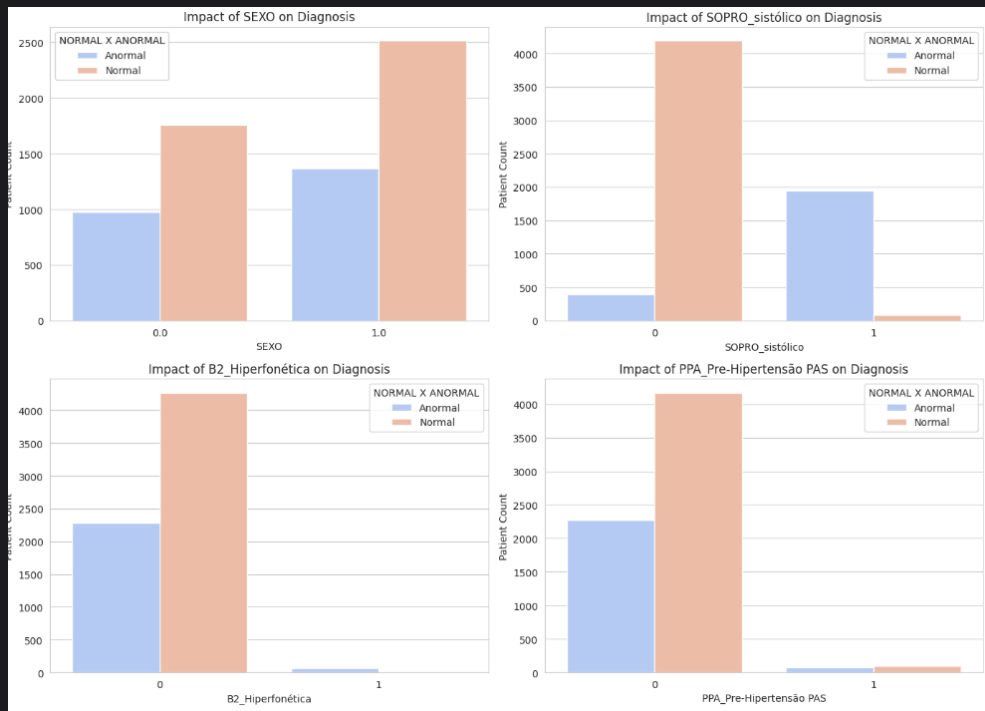
Basic Description - Demographics



Basic Description - Demographics



Bivariate Analysis



Multivariate Analysis I (Patient Clusters)

Method: K-Means Clustering (k=3) + PCA Projection

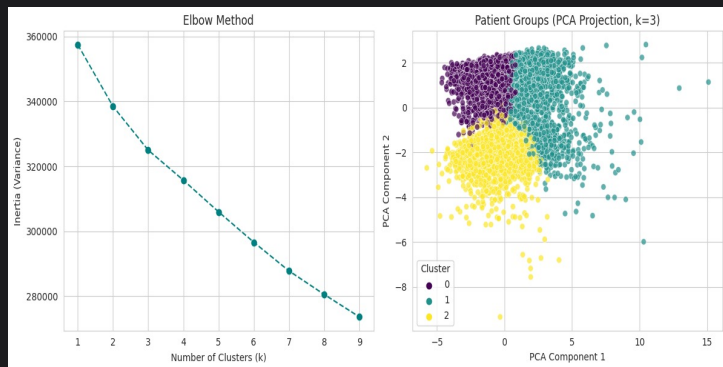
Identified Profiles:

Cluster 0 (Young & Healthy): Low pathology rate.

Cluster 1 (Adolescents): Physiological higher BP, but lowest risk.

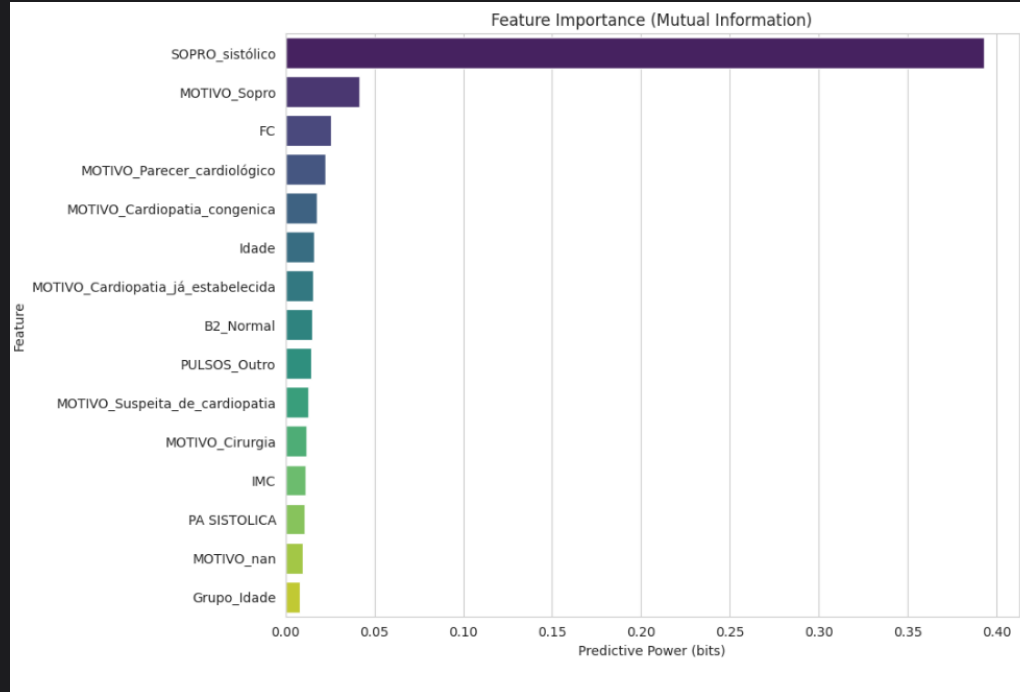
Cluster 2 (Symptomatic): Defined by Systolic Murmurs and high pathology rate (55%).

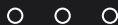
Conclusion: Patients are stratified by Growth Stage and Clinical Symptoms.



	Idade	IMC	PA SISTOLICA	PA DIASTOLICA	target	SOPRO_sistólico	PPA_Pre-Hipertensão PAS
Cluster							
0	6.339839	17.013452	96.184721	58.665724	0.282767	0.227971	0.016572
1	12.743717	21.203116	108.602081	67.303271	0.229011	0.179881	0.036174
2	7.332686	17.335864	98.723344	60.643964	0.555712	0.517614	0.026303

Multivariate Analysis II (Feature Importance)





Predictive Models (Supervised Learning)

Algorithms: Decision Tree, SVM, Random Forest, Naive Bayes.

Best Performer: Random Forest.

Accuracy: ~92.8% | ROC-AUC: 0.94.

Sensitivity: ~83%

Success Factor: Successfully leveraged the "Murmur" variable to identify high-risk patients.



Predictive Models (Supervised Learning)

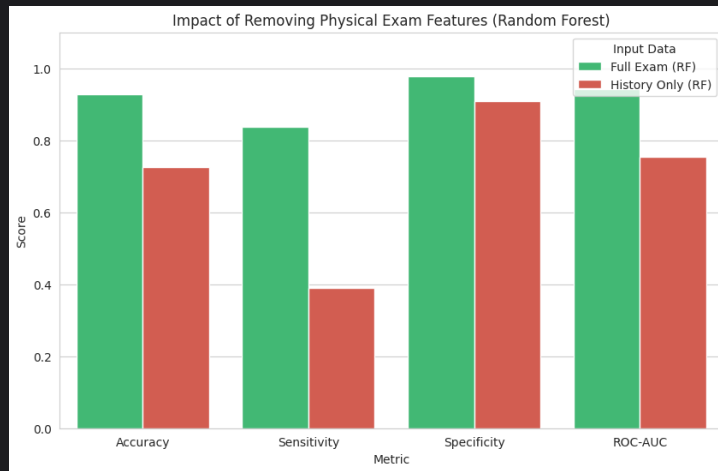
Objective: Test if AI can diagnose without the physical exam.

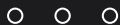
The Experiment: Trained "History Only" models (Age, BMI, Heart Rate only).

Results:

Sensitivity Collapse: Detection rate dropped from ~83% to <40%.

Implication: A history-only tool would miss 60% of sick children.





Discussion & Conclusion

Main Conclusion: The "Physical Exam" (Murmur) is mathematically indispensable.

Model Role:

- Decision Trees: Best for explainability.
- Random Forest: Best for raw accuracy.

Final Recommendation: Deploy models as Triage Assistants to process exam findings, not as standalone screening tools.

